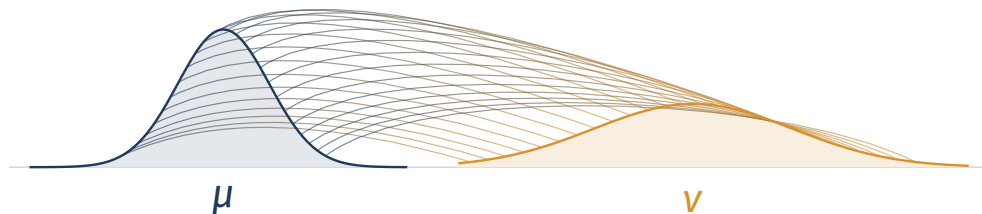


Introduction to Optimal Transportation

From moving sand to the geometry of probability measures

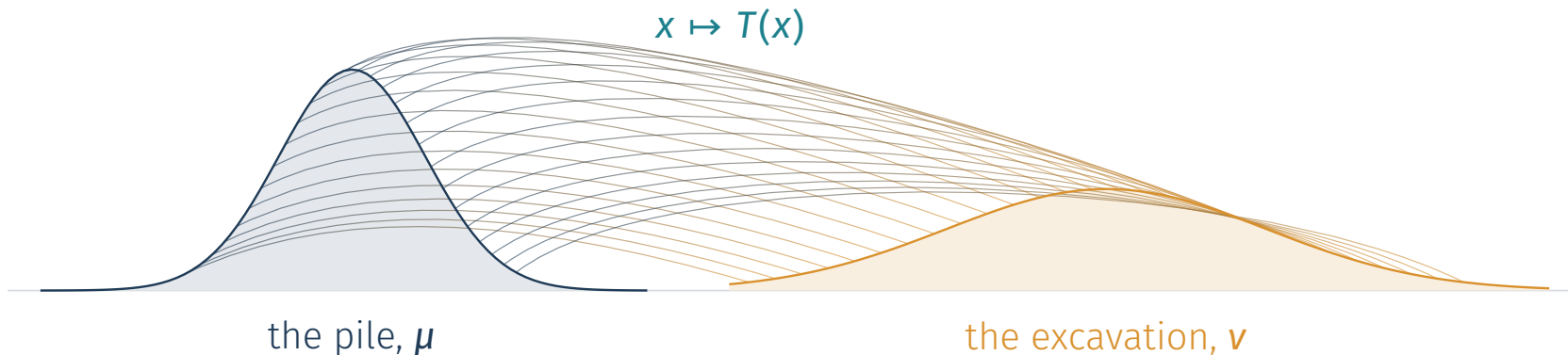
Vardan Voskanyan

University of Coimbra, Portugal



Monge's problem, 1781

Gaspard Monge: move a pile of earth into an excavation of the same volume, at the least total cost. Moving a grain from x to $T(x)$ costs $c(x, T(x))$.



Which assignment $x \mapsto T(x)$ costs the least?

Three questions

1. **Existence.** Is there an optimal way to move the mass at all?

Monge's formulation resists direct attack; Kantorovich's relaxation answers it.

2. **Structure.** What does the optimal map look like?

Duality and convex analysis reveal it; for the quadratic cost, a gradient of a convex function.

3. **Geometry.** What does transport say about the space of measures?

The optimal cost is a distance; geodesics, convexity and gradient flows follow.

Along the way: the isoperimetric and Brunn–Minkowski inequalities, proved by transport.

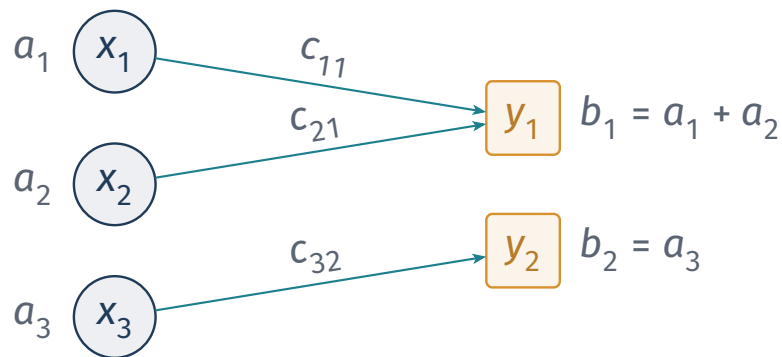
Monge's formulation

From mines and factories to measures. Why the problem is hard.

	amounts	locations
n mines	$a_1, \dots, a_n$	$x_1, \dots, x_n$
m factories	$b_1, \dots, b_m$	$y_1, \dots, y_m$

Supply equals demand: $\sum_i a_i = \sum_j b_j$.

Moving one unit from x_i to y_j costs $c_{ij} := c(x_i, y_j)$.



a transport map: each mine serves one factory

Transport maps

A **transport map** $T : \{x_1, \dots, x_n\} \rightarrow \{y_1, \dots, y_m\}$ sends each mine to one factory and meets every demand:

$$b_j = \sum_{i: T(x_i)=y_j} a_i \quad \text{for all } j.$$

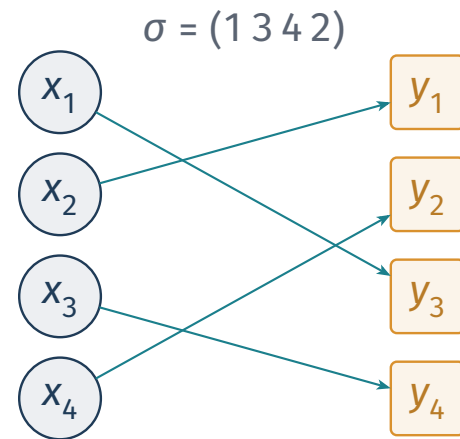
Monge's problem, discrete version: among all such maps,

$$\sum_{i=1}^n a_i c(x_i, T(x_i)) \longrightarrow \min.$$

$m = n$, all $a_i = b_j = 1/n$. A transport map is a bijection, a permutation $\sigma \in S_n$:

$$\min_{\sigma \in S_n} \frac{1}{n} \sum_{i=1}^n c(x_i, y_{\sigma(i)}).$$

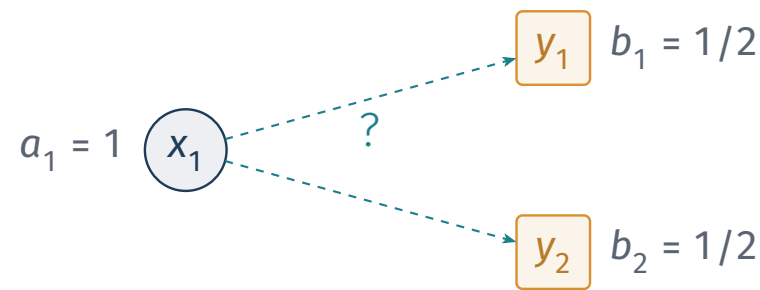
Finite, but with $n!$ candidates. The structure we develop later (duality) is what algorithms use.



A map cannot split the mass of a mine.

One mine, two factories: $a_1 = 1$, $b_1 = b_2 = 1/2$. No admissible map.

The constraint $T_{\#}\mu = \nu$ can be empty, and it is not closed under limits. We will need a more flexible notion of moving mass.



From sums to measures

With δ_x the unit mass at x , mines and factories are probability measures and the cost of a map is an integral:

$$\mu = \sum_{i=1}^n a_i \delta_{x_i}, \quad \nu = \sum_{j=1}^m b_j \delta_{y_j},$$

$$\sum_i a_i c(x_i, T(x_i)) = \int_X c(x, T(x)) d\mu(x).$$

STANDING ASSUMPTIONS

X, Y Polish spaces (think: regions of $\mathbb{R}^n$); $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$; a cost $c : X \times Y \rightarrow [0, +\infty]$, e.g. $|x - y|^p$.

Push-forward: what *sending* μ to ν means

DEFINITION

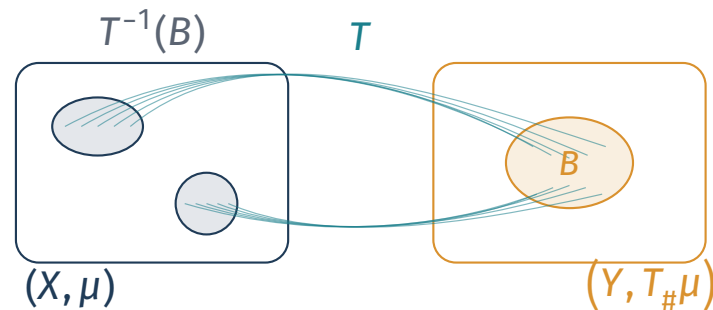
For $\mu \in \mathcal{P}(X)$ and a Borel map $T : X \rightarrow Y$, the push-forward $T_{\#}\mu \in \mathcal{P}(Y)$ is

$$T_{\#}\mu(B) := \mu(T^{-1}(B)) \quad \text{for all Borel } B \subset Y.$$

The mass $T_{\#}\mu$ sees in B is the mass μ had in the region T sends into B .

For every test function h ,

$$\int_Y h \, d(T_{\#}\mu) = \int_X h \circ T \, d\mu.$$



The Monge problem

MONGE PROBLEM

$$(MP) \quad \inf \left\{ \int_X c(x, T(x)) d\mu(x) : T_{\#}\mu = \nu \right\}.$$

For densities $d\mu = f dx$, $d\nu = g dy$ and a diffeomorphism T , the constraint is a PDE,

$$f(x) = g(T(x)) | \det \nabla T(x) |,$$

Fully non-linear, of Monge–Ampère type. The admissible set is neither convex nor weakly closed, so the direct method fails.

Where we stand



Monge's problem is natural but rigid:

- admissible maps may not exist (mass cannot be split);
- minimising sequences of maps need not converge to a map.

Kantorovich, 1942: allow mass to split. The problem becomes linear.

Kantorovich relaxation

Split the mass, linearise the problem, and existence follows.

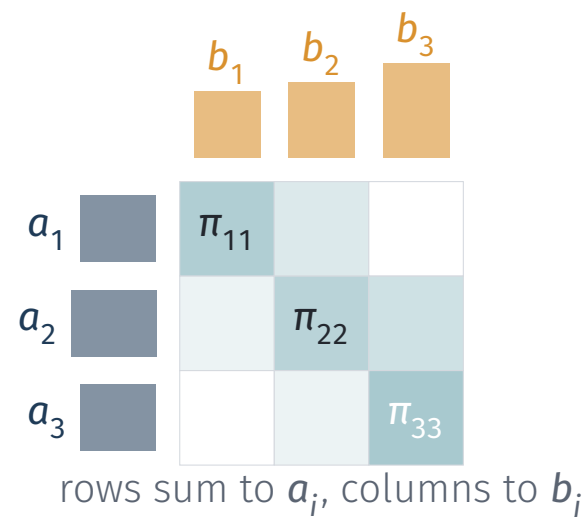
Transport plans: the discrete picture

Prescribe **how much** mass goes from each source to each destination: a matrix $\pi = (\pi_{ij}) \geq 0$ with

$$\sum_j \pi_{ij} = a_i, \quad \sum_i \pi_{ij} = b_j.$$

Cost: $\sum_{i,j} \pi_{ij} c(x_i, y_j)$, linear in π .

A linear program on a compact convex set: a minimiser exists.



Transport plans: general measures

The matrix is a measure on the product, $\pi = \sum_{ij} \pi_{ij} \delta_{(x_i, y_j)}$, with cost $\int_{X \times Y} c \, d\pi$.

DEFINITION

A *transport plan* is $\pi \in \mathcal{P}(X \times Y)$ with marginals μ and ν :

$$\pi(A \times Y) = \mu(A), \quad \pi(X \times B) = \nu(B).$$

$\Pi(\mu, \nu)$ denotes the set of transport plans.

$d\pi(x, y)$ is the mass moved from dx to dy . The constraints are *linear* tests:

$$\int_{X \times Y} (\varphi(x) + \psi(y)) \, d\pi = \int_X \varphi \, d\mu + \int_Y \psi \, d\nu \quad \forall \varphi \in C_b(X), \psi \in C_b(Y).$$

The Kantorovich problem

KANTOROVICH PROBLEM

$$(KP) \quad T_c(\mu, \nu) := \inf \left\{ I_c[\pi] := \int_{X \times Y} c \, d\pi : \pi \in \Pi(\mu, \nu) \right\}.$$

- **Plans always exist:** $\mu \otimes \nu \in \Pi(\mu, \nu)$.
- **Every map is a plan:** $\pi_T := (\text{Id}, T)_\# \mu$ has cost $\int c(x, T(x)) \, d\mu$, so $(KP) \leq (MP)$.
- **Linear objective, linear constraints:** a convex problem. Existence and duality become available.

Do we lose anything by relaxing?

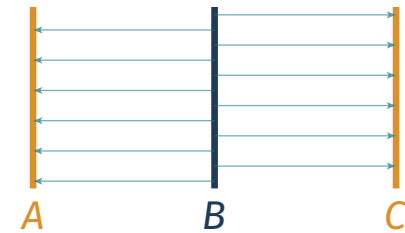
Counterexample: a map cannot be optimal

Setup. Unit segments A, B, C at $x = -1, 0, 1$ with length measures $\mathcal{L}_A, \mathcal{L}_B, \mathcal{L}_C$; cost $c = |x - y|^p$. Move the middle one onto the outer two, half each:

$$\mu = \mathcal{L}_B, \quad \nu = \frac{1}{2}(\mathcal{L}_A + \mathcal{L}_C).$$

Claim: no map is optimal. Every point of B travels at least 1, so every map costs ≥ 1 . Cost 1 forces $T(0, t) = (\pm 1, t)$, but such a map fills only length 1 of $A \cup C$ at full density, while ν spreads half density over length 2.

a near-optimal map: alternate left / right



Yet the infimum over maps is 1. Cut B into $2n$ pieces, A and C into n each; send even pieces left, odd right:

$$T_n : B_{2i} \rightarrow A_i, \quad B_{2i-1} \rightarrow C_i, \quad (T_n)_\# \mu = \nu, \quad \int |x - T_n(x)|^p d\mu \rightarrow 1.$$

The limit is a plan, not a map. T_n oscillates ever faster, but the induced plans converge weakly to the optimal plan, which splits every point of B half left, half right:

$$(\text{Id}, T_n)_\# \mu \rightarrow \pi = \int_0^1 \frac{1}{2} (\delta_{(0,t),(-1,t)} + \delta_{(0,t),(1,t)}) dt.$$

Kantorovich's problem is the closure of Monge's: for continuous c and atomless μ , $(\text{KP}) = (\text{MP})$ even when (MP) is not attained.

Compactness in the space of measures

DEFINITION

$\mathcal{F} \subset \mathcal{P}(X)$ is *tight* if for every $\varepsilon > 0$ there is a compact K_ε with $\mu(X \setminus K_\varepsilon) < \varepsilon$ for all $\mu \in \mathcal{F}$.

THEOREM Prokhorov

A tight sequence in $\mathcal{P}(X)$ has a weakly convergent subsequence, $\int \varphi d\mu_{n_k} \rightarrow \int \varphi d\mu$ for all $\varphi \in C_b(X)$. On a Polish space the converse holds too.

Single measures on Polish spaces are tight, so $\Pi(\mu, \nu)$ is tight: use $K_\varepsilon \times K'_\varepsilon$.

THEOREM

Let X, Y be Polish and $c : X \times Y \rightarrow [0, +\infty]$ lower semi-continuous. Then (KP) has a minimiser.

1. **Minimising sequence.** $\pi_n \in \Pi(\mu, \nu)$, $I_c[\pi_n] \rightarrow T_c(\mu, \nu)$.
2. **Compactness.** $\Pi(\mu, \nu)$ is tight, so $\pi_{n_k} \rightarrow \pi$.
3. **Admissibility.** Marginal constraints are linear tests against C_b functions: they pass to the limit.
4. **Semi-continuity.** $c = \sup_k c_k$ with c_k bounded continuous, so $I_c = \sup_k I_{c_k}$ is weakly l.s.c. and $I_c[\pi] \leq \liminf_k I_c[\pi_{n_k}] = T_c(\mu, \nu)$. ■

Where we stand



Existence is settled for the relaxed problem: optimal plans exist for every l.s.c. cost.

We know nothing yet about *what* they look like, or whether they are maps.

Next: the second gift of linearity, duality.

Kantorovich duality

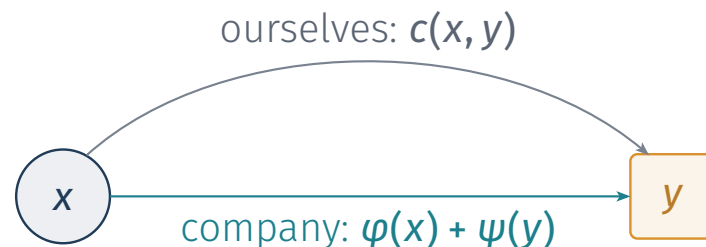
Prices instead of routes. The dual problem reveals the structure of optimal plans.

Hire a company instead of moving the goods. It charges $\varphi(x)$ per unit *collected* at x and $\psi(y)$ per unit *delivered* at y .

We accept only if it never beats doing it ourselves:

$$\varphi(x) + \psi(y) \leq c(x, y) \quad \forall (x, y).$$

The company maximises its income over all admissible price systems.



The dual problem, and weak duality

DUAL PROBLEM

$$(DP) \quad \sup \left\{ J(\varphi, \psi) := \int_X \varphi \, d\mu + \int_Y \psi \, d\nu : \varphi \oplus \psi \leq c \right\}.$$

For any $\pi \in \Pi(\mu, \nu)$ and admissible (φ, ψ) ,

$$J(\varphi, \psi) = \int_{X \times Y} (\varphi(x) + \psi(y)) \, d\pi \leq \int_{X \times Y} c \, d\pi = I_c[\pi],$$

so $(DP) \leq (KP)$: weak duality.

The real content of duality is the reverse inequality: prices can be chosen so that no money is lost.

Deriving the dual: Lagrange multipliers

Penalise the marginal constraints with multipliers (φ, ψ) :

$$\inf_{\pi \in \Pi(\mu, \nu)} I_c[\pi] = \inf_{\pi \geq 0} \sup_{\varphi, \psi} \left\{ \int c \, d\pi - \int (\varphi(x) + \psi(y)) \, d\pi + \int \varphi \, d\mu + \int \psi \, d\nu \right\}.$$

The inner supremum is $+\infty$ if π violates a constraint (scale the offending φ or ψ), and $I_c[\pi]$ otherwise. Nothing has changed yet.

Deriving the dual: swap inf and sup

$$\sup_{\varphi, \psi} \left\{ \int \varphi \, d\mu + \int \psi \, d\nu + \underbrace{\inf_{\pi \geq 0} \int (c - \varphi \oplus \psi) \, d\pi}_{(*)} \right\}.$$

- If $c - \varphi \oplus \psi < 0$ at some (x_0, y_0) : take $\pi = \lambda \delta_{(x_0, y_0)}$, $\lambda \rightarrow \infty$, so $(*) = -\infty$.
- If $\varphi \oplus \psi \leq c$ everywhere: $\pi = 0$ gives $(*) = 0$.

Only admissible pairs survive, and the bracket is $J(\varphi, \psi)$: this is (DP).

$\inf \sup = \sup \inf$ is the one non-rigorous step. Convex duality (Fenchel–Rockafellar) justifies it.

Kantorovich duality

THEOREM

X, Y Polish, $c \geq 0$ l.s.c. Let Φ_c be the admissible pairs in $L^1(\mu) \times L^1(\nu)$ and A_c those in $C_b(X) \times C_b(Y)$. Then

$$\inf_{\pi \in \Pi(\mu, \nu)} I_c[\pi] = \sup_{\Phi_c} J(\varphi, \psi) = \sup_{A_c} J(\varphi, \psi).$$

Strong duality holds without any compactness: only lower semi-continuity of the cost is used.

Three problems, one value

Monge	(MP)	$\inf\{\int c(x, T(x)) d\mu : T_{\#}\mu = \nu\}$	maps
Kantorovich	(KP)	$\inf\{\int c d\pi : \pi \in \Pi(\mu, \nu)\}$	plans
Dual	(DP)	$\sup\{\int \varphi d\mu + \int \psi d\nu : \varphi \oplus \psi \leq c\}$	prices

$$(DP) = (KP) \leq (MP)$$

The primal tells us where mass goes; the dual tells us what it is worth. The dual is the one we can compute with.

Relabel so that the identity is optimal: $\sum_i c_{ii} \leq \sum_i c_{i\sigma(i)}$ for all σ . We need φ_i, ψ_j with

$$\varphi_i + \psi_j \leq c_{ij} \quad \forall i, j, \quad \varphi_i + \psi_i = c_{ii} \quad \forall i.$$

Eliminate $\psi_j = c_{ji} - \varphi_i$. With $b_{ij} := c_{ij} - c_{ji}$ the task becomes

$$\varphi_i - \varphi_j \leq b_{ij}, \quad b_{ii} = 0, \quad \sum_i b_{i\sigma(i)} \geq 0 \quad \forall \sigma.$$

Difference constraints: shortest paths in a graph with edge weights b_{ij} .

Fix $\varphi_1 = 0$. Along any path $i = i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_k = 1$,

$$\varphi_i \leq b_{ii_1} + b_{i_1i_2} + \dots + b_{i_{k-1}1}.$$

Define φ_i as the minimum over all paths from i to 1 . No negative cycles ($\sum_i b_{i\sigma(i)} \geq 0$) makes it finite, and $\varphi_i - \varphi_j \leq b_{ij}$ is the triangle inequality for shortest paths.

Remember this: the potential is an infimum over chains of costs. The same formula returns in the continuous theory.

Kantorovich duality: from prices to maps



We have the dual problem and know it is exact. The next slides extract structure from it: **c**-concave potentials, cyclical monotonicity, and finally a formula for the map.

The c-transform

$$\varphi^c(y) := \inf_{x \in X} \{c(x, y) - \varphi(x)\}, \quad \psi^c(x) := \inf_{y \in Y} \{c(x, y) - \psi(y)\}.$$

φ^c is the *largest* ψ making (φ, ψ) admissible.

DEFINITION

φ is *c-concave* if it is a *c*-transform. Always $\varphi \leq \varphi^{cc}$; φ is *c-concave* iff $\varphi = \varphi^{cc}$; $(\varphi^{cc})^c = \varphi^c$.

Improving a dual pair: from $\psi \leq \varphi^c$ and $\varphi \leq \varphi^{cc}$,

$$J(\varphi, \psi) \leq J(\varphi, \varphi^c) \leq J(\varphi^{cc}, \varphi^c).$$

Why c -concave functions: regularity for free

An infimum of the functions $y \mapsto c(x, y) - \varphi(x)$ inherits the modulus of continuity of c : if c is Lipschitz on a compact set, all c -concave functions are uniformly Lipschitz.

CONSEQUENCE

A maximising sequence (φ_k, φ_k^c) is equi-continuous and, after subtracting constants, equi-bounded. By Arzelà–Ascoli a subsequence converges uniformly to a maximiser (φ, φ^c) of (DP).

Such a φ is a **Kantorovich potential**. The dual is attained, while the primal only gave a plan.

LEGENDRE TRANSFORM

$$f^*(y) := \sup_x \{x \cdot y - f(x)\}$$

is convex l.s.c., and $f^{**} = f$
iff f is convex l.s.c.

SUBDIFFERENTIAL

$$\partial f(x) := \{p : f(z) \geq f(x) + p \cdot (z - x) \forall z\}$$

$$y \in \partial f(x) \iff x \in \partial f^*(y) \iff f(x) + f^*(y) = x \cdot y$$

For $c(x, y) = \frac{1}{2} |x - y|^2$ expand the square in φ^c and set $\tilde{\varphi} := \frac{1}{2} |x|^2 - \varphi$, $\tilde{\psi} := \frac{1}{2} |y|^2 - \varphi^c$:

$$\tilde{\psi} = \tilde{\varphi}^*, \quad \text{so} \quad \varphi \text{ is } c\text{-concave} \iff \frac{1}{2} |x|^2 - \varphi(x) \text{ is convex.}$$

Cyclical monotonicity

DEFINITION

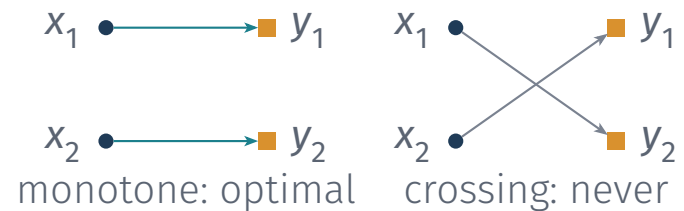
$\Gamma \subset \mathbb{R}^n \times \mathbb{R}^n$ is *cyclically monotone* if for all $(x_i, y_i)_{i=1}^N \subset \Gamma$ and $\sigma \in S_N$

$$\sum_{i=1}^N x_i \cdot y_i \geq \sum_{i=1}^N x_i \cdot y_{\sigma(i)}.$$

$N = 2$: $(x_1 - x_2) \cdot (y_1 - y_2) \geq 0$, ordinary monotonicity.

$\frac{1}{2} \|x - y\|^2 = \frac{1}{2} \|x\|^2 + \frac{1}{2} \|y\|^2 - x \cdot y$ and the squares are fixed by the marginals, so

maximising $\sum x_i \cdot y_i$ is minimising the quadratic cost.



Cyclically monotone sets are subdifferentials

The graph $\Gamma_f := \{(x, y) : y \in \partial f(x)\}$ of a convex f is cyclically monotone: sum $f(x_{i+1}) - f(x_i) \geq y_i \cdot (x_{i+1} - x_i)$ around a cycle; the left side telescopes to 0.

THEOREM Rockafellar

Every cyclically monotone set is contained in the subdifferential graph of a convex l.s.c. function.

The road to Brenier: optimality $\Rightarrow$ cyclical monotonicity $\Rightarrow$ the plan lives on $\partial f \Rightarrow$ where f is differentiable, $y = \nabla f(x)$.

General costs: c -cyclical monotonicity

DEFINITION

$\Gamma \subset X \times Y$ is c -cyclically monotone if for all $(x_i, y_i)_{i=1}^N \subset \Gamma$ and $\sigma \in S_N$

$$\sum_{i=1}^N c(x_i, y_i) \leq \sum_{i=1}^N c(x_i, y_{\sigma(i)}).$$

THEOREM Rockafellar–Rüschendorf

If $\Gamma \neq \emptyset$ is c -cyclically monotone, there is a c -concave φ with $\Gamma \subset \{(x, y) : \varphi(x) + \varphi^c(y) = c(x, y)\}$.

For $c(x, y) = -x \cdot y$ this is Rockafellar's theorem with $\varphi = -f$.

The potential is a shortest-path distance

Fix $(x_0, y_0) \in \Gamma$ and take the infimum over all chains $(x_1, y_1), \dots, (x_k, y_k) \in \Gamma$:

$$\varphi(x) := \inf\{[c(x, y_k) - c(x_k, y_k)] + \dots + [c(x_1, y_0) - c(x_0, y_0)]\}.$$

- c -cyclical monotonicity: closed chains have non-negative cost, so $\varphi > -\infty$.
- Prepending $(x, y) \in \Gamma$ to a chain gives $\varphi(x') \leq c(x', y) - c(x, y) + \varphi(x)$, i.e. $\varphi(x) + \varphi^c(y) \geq c(x, y)$.

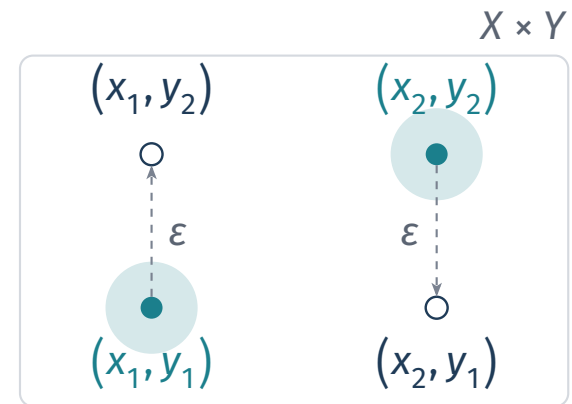
The shortest-path potential of the assignment problem, with b_{ij} replaced by $c(x_{j+1}, y_j) - c(x_j, y_j)$.

Optimal plans are c -cyclically monotone

THEOREM

Let c be continuous. π is optimal for (KP) $\iff \text{supp } \pi$ is c -cyclically monotone.

Proof idea ($\Rightarrow$). If points $(x_i, y_i) \in \text{supp } \pi$ and σ violate the inequality, move a little mass ε from balls around (x_i, y_i) to balls around $(x_i, y_{\sigma(i)})$. Marginals are unchanged; the cost strictly drops.



From the potential to the map

If π is optimal and φ a Kantorovich potential, duality gives $\int c \, d\pi = \int (\varphi \oplus \varphi^c) \, d\pi$ with $\varphi \oplus \varphi^c \leq c$, so

$$\text{supp } \pi \subset \Gamma_\varphi := \{(x, y) : \varphi(x) + \varphi^c(y) = c(x, y)\}.$$

On Γ_φ the function $x \mapsto c(x, y_0) - \varphi(x)$ is minimal at x_0 . Where things are differentiable,

$$\nabla_x c(x_0, y_0) = \nabla \varphi(x_0).$$

If $y \mapsto \nabla_x c(x, y)$ is injective (the twist condition), this determines $y_0 = T(x_0)$: the plan is a map.

Where we stand



A recipe for the structure of optimal plans:

1. find a Kantorovich potential φ (Arzelà–Ascoli);
2. the plan lives where $\varphi(x) + \varphi^c(y) = c(x, y)$;
3. differentiate, $\nabla_x c(x, y) = \nabla \varphi(x)$, and solve for y .

Step 3 depends on the cost: convex, concave, linear.

The structure of optimal maps

Convex costs give maps, the quadratic cost gives gradients of convex functions.

Strictly convex costs

$c(x, y) = h(x - y)$ with h strictly convex: $\nabla_x c = \nabla h(x - y)$, and ∇h is invertible with inverse ∇h^* .

THEOREM Gangbo–McCann

If μ is absolutely continuous, the optimal plan is unique and induced by the map

$$T(x) = x - \nabla h^*(\nabla \varphi(x)), \quad \varphi \text{ a Kantorovich potential.}$$

In particular $(\text{KP}) = (\text{MP})$.

Why absolutely continuous: c -concave functions are locally Lipschitz, hence differentiable a.e. (Rademacher), so $\nabla \varphi$ exists μ -a.e.

The quadratic cost: Brenier's theorem

THEOREM Brenier, 1991

$c(x, y) = \frac{1}{2} \|x - y\|^2$ on $\mathbb{R}^n$, μ, ν with finite second moments, μ absolutely continuous. The optimal plan is unique and induced by the gradient of a convex function:

$$T(x) = x - \nabla \varphi(x) = \nabla \left(\frac{1}{2} \|x\|^2 - \varphi(x) \right) = \nabla \tilde{\varphi}(x), \quad \tilde{\varphi} \text{ convex.}$$

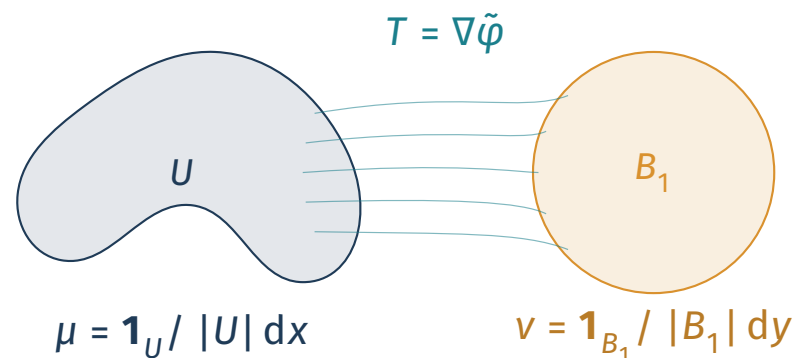
Conversely, a gradient of a convex function pushing μ to ν is the optimal map.

$h = \frac{1}{2} \|\cdot\|^2$ gives $\nabla h^* = \text{Id}$; convexity of $\tilde{\varphi}$ is the dictionary. Brenier's map is the multidimensional monotone rearrangement.

Among sets of given volume the ball has least perimeter. By scaling,

$$\text{Per}(U) \geq n |B_1|^{1/n} |U|^{1-1/n}.$$

Idea: push the uniform measure on U to the uniform measure on B_1 with Brenier's map, and use only that $\nabla T \succeq 0$ and $|T| \leq 1$.



1. Change of variables. $T_{\#}\mu = \nu$ reads $\det \nabla T = |B_1| / |U|$ a.e. in U .

2. Arithmetic–geometric mean. $\nabla T = \nabla^2 \tilde{\varphi} \succeq 0$, so on its eigenvalues

$$\left(\frac{|B_1|}{|U|} \right)^{1/n} = (\det \nabla T)^{1/n} \leq \frac{1}{n} \operatorname{tr} \nabla T = \frac{1}{n} \operatorname{div} T.$$

3. Integrate. Divergence theorem and $|T| \leq 1$:

$$|B_1|^{1/n} |U|^{1-1/n} \leq \frac{1}{n} \int_U \operatorname{div} T \, dx = \frac{1}{n} \int_{\partial U} T \cdot \vec{n} \, d\mathcal{H}^{n-1} \leq \frac{1}{n} \operatorname{Per}(U). \quad \blacksquare$$

Strictly concave costs

$c(x, y) = h(|x - y|)$ with h strictly concave, e.g. $|x - y|^p$, $0 < p < 1$. Concave costs favour *not moving* mass already in place: one particle over $2r$ is cheaper than two over r . Split off the common part,

$$\mu \wedge \nu := \min\{\mu, \nu\}, \quad (\mu - \nu)_+ := \mu - \mu \wedge \nu.$$

THEOREM Gangbo–McCann

$h \in C^1$ strictly concave, $h(0) = 0$, $(\mu - \nu)_+$ absolutely continuous. The optimal plan is unique, fixes the common mass and moves the rest by a map:

$$\pi = (\text{Id}, \text{Id})_{\#}(\mu \wedge \nu) + (\text{Id}, T)_{\#}(\mu - \nu)_+.$$

The distance cost $c(x, y) = |x - y|$

Neither strictly convex nor concave; the twist condition fails.

- **Non-uniqueness.** In 1D with $\text{supp } \mu$ left of $\text{supp } \nu$, every plan is optimal: the cost is $\int y \, d\nu - \int x \, d\mu$.
- **Transport rays.** Potentials are 1-Lipschitz and $\varphi(x) - \varphi(y) = |x - y|$ on the contact set: mass moves along segments where φ has slope 1.
- **Selection.** Among optimal plans, the one minimising a strictly convex secondary cost is unique and is a map, monotone on each ray.

Strictly convex: maps and uniqueness. Strictly concave: maps off the common mass.
Linear: rays and selection.

Where we stand



Structure is settled. For strictly convex costs the optimal plan is a map, $T = x - \nabla h^*(\nabla \varphi)$; for the quadratic cost, the gradient of a convex function.

Next: use the optimal cost as a distance between measures, and see what geometry it induces.

Wasserstein geometry

The optimal cost is a distance. Geodesics, convexity and gradient flows on the space of measures.

The Wasserstein distance

On a Polish metric space (X, d) , for $p \geq 1$ and $\mathcal{P}_p(X) := \{\mu : \int d(x, x_0)^p d\mu < \infty\}$,

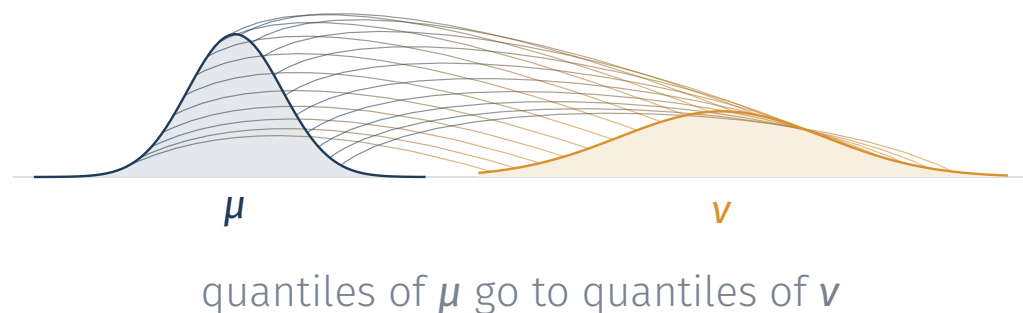
$$W_p(\mu, \nu)^p := \inf \left\{ \int_{X \times X} d(x, y)^p d\pi : \pi \in \Pi(\mu, \nu) \right\}.$$

- W_p is a metric: the triangle inequality comes from *gluing* plans along a common marginal.
- $W_p(\mu_n, \mu) \rightarrow 0$ iff $\mu_n \rightarrow \mu$ and p -th moments converge: W_p metrises weak convergence on bounded sets.
- $W_p(\delta_x, \delta_y) = d(x, y)$: X embeds isometrically in $\mathcal{P}_p(X)$.

With distribution functions $F_\mu(x) = \mu((-\infty, x])$ and quantile functions $F_\mu^{-1}(t) = \inf\{x : F_\mu(x) \geq t\}$,

$$W_p(\mu, \nu)^p = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^p dt.$$

The optimal plan is $(F_\mu^{-1}, F_\nu^{-1})_\# \mathcal{L}^1|_{[0,1]}$. If μ has no atoms, the optimal map is the **monotone rearrangement** $T = F_\nu^{-1} \circ F_\mu$, for every $p > 1$ at once.



For $\mu = N(m_\mu, \Sigma_\mu)$, $\nu = N(m_\nu, \Sigma_\nu)$ the optimal map for the quadratic cost is affine, and

$$W_2(\mu, \nu)^2 = |m_\mu - m_\nu|^2 + \text{tr}\left(\Sigma_\mu + \Sigma_\nu - 2\left(\Sigma_\mu^{1/2}\Sigma_\nu\Sigma_\mu^{1/2}\right)^{1/2}\right).$$

Isotropic case $\Sigma = \sigma^2 \text{Id}$ (or $n = 1$):

$$W_2(\mu, \nu)^2 = |m_\mu - m_\nu|^2 + n |\sigma_\mu - \sigma_\nu|^2, \quad T(x) = m_\nu + \frac{\sigma_\nu}{\sigma_\mu}(x - m_\mu).$$

$T = \nabla \tilde{\varphi}$ with $\tilde{\varphi}(x) = m_\nu \cdot x + \frac{\sigma_\nu}{2\sigma_\mu} |x - m_\mu|^2$ convex, as Brenier predicts.

$p = 1$: the Kantorovich–Rubinstein distance

$$W_1(\mu, \nu) = \sup \left\{ \int_X u \, d(\mu - \nu) : \text{Lip}(u) \leq 1 \right\}.$$

Why one function suffices: restrict the dual to pairs $(\varphi^d, \varphi^{dd})$ with $c = d$.

- $\varphi^d = \inf_x \{d(x, \cdot) - \varphi(x)\}$ is 1-Lipschitz;
- for 1-Lipschitz u , $u^d(x) = \inf_y \{d(x, y) - u(y)\} = -u(x)$.

So the pair is $(u, -u)$ and $J(u, -u) = \int u \, d(\mu - \nu)$.

W_1 depends on $\mu - \nu$ only. Used in the Wasserstein GAN and in concentration inequalities.

Wasserstein geometry: curves and geodesics



W_p makes $\mathcal{P}_p$ a metric space. We now study curves in it, identify their speed with a velocity field, and find the geodesics: displacement interpolation.

Curves of measures

DEFINITION

$\rho : [0, 1] \rightarrow \mathcal{P}_p(\mathbb{R}^n)$ is *absolutely continuous* if $W_p(\rho_s, \rho_t) \leq \int_s^t g$ for some $g \in L^1$. Its *metric derivative*

$$|\rho'| (t) := \lim_{h \rightarrow 0} \frac{W_p(\rho_{t+h}, \rho_t)}{|h|}$$

exists a.e. and is the smallest such g .

Measures do not subtract, so a curve of measures has no velocity. The metric derivative is its speed; we want to see it as a velocity field moving particles.

Moving measures with a vector field

A vector field $v : [0, 1] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ moves particles, $\dot{X}(t, x) = v(t, X(t, x))$, $X(0, x) = x$, and hence measures: $\rho_t := X(t, \cdot)_\# \rho_0$.

CONTINUITY EQUATION

$$\partial_t \rho_t + \operatorname{div}(v_t \rho_t) = 0, \quad \text{weakly:} \quad \frac{d}{dt} \int \varphi \, d\rho_t = \int \nabla \varphi \cdot v_t \, d\rho_t \quad \forall \varphi \in C_c^1.$$

Proof. $\frac{d}{dt} \int \varphi(X(t, x)) \, d\rho_0 = \int \nabla \varphi(X) \cdot v(t, X) \, d\rho_0 = \int \nabla \varphi \cdot v_t \, d\rho_t.$ ■

Absolutely continuous curves solve the continuity equation

THEOREM Ambrosio–Gigli–Savaré

Let $p > 1$.

1. If (ρ_t, v_t) solves the continuity equation with $\int_0^1 \|v_t\|_{L^p(\rho_t)} dt < \infty$, then ρ is absolutely continuous and $|\rho'| (t) \leq \|v_t\|_{L^p(\rho_t)}$.
2. Conversely, every absolutely continuous curve admits a velocity field with $|\rho'| (t) = \|v_t\|_{L^p(\rho_t)}$ a.e. For $p = 2$ it lies in the closure of gradients in $L^2(\rho_t)$.

Metric speed = norm of the optimal velocity field. This is the bridge between W_p and PDE.

Geodesics: displacement interpolation

A **constant-speed geodesic** satisfies $W_p(\rho_s, \rho_t) = |t - s| W_p(\rho_0, \rho_1)$.

THEOREM McCann

$p > 1$, μ absolutely continuous, T the optimal map. Then

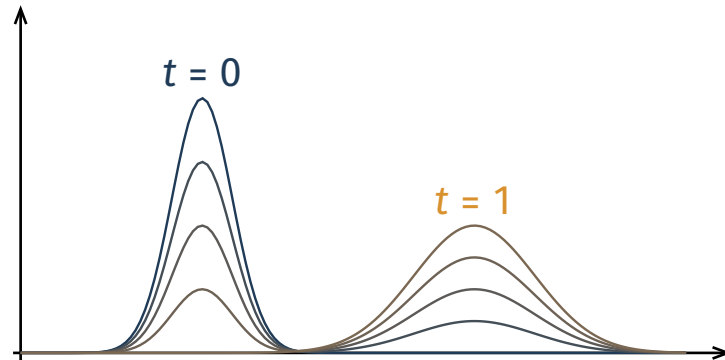
$$T_t := (1 - t)\text{Id} + tT, \quad \rho_t := (T_t)_\# \mu$$

is the constant-speed geodesic from μ to ν . In general, $\rho_t = ((1 - t)x + ty)_\# \pi$ for an optimal plan π .

$(T_s, T_t)_\# \mu$ is a plan of cost $|t - s|^p W_p^p$, so $W_p(\rho_s, \rho_t) \leq |t - s| W_p(\mu, \nu)$, and the triangle inequality forces equality.

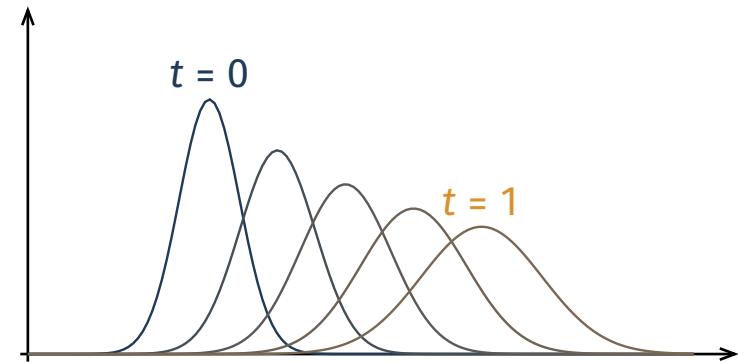
Five snapshots, $t = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$, of two curves from μ (navy) to ν (amber).

linear: $(1 - t)\mu + t\nu$



Mass appears at the target and disappears at the source; the intermediate measures are bimodal. Nothing moves.

displacement: $\rho_t = (T_t)_\# \mu$



Mass travels. Each intermediate measure is a translated, rescaled Gaussian. This is the W_2 geodesic.

The Benamou–Brenier formula

Along the geodesic, $v(t, x) = (T - \text{Id})(T_t^{-1}(x))$ is constant along trajectories and $\int_0^1 \int |v_t|^p d\rho_t dt = W_p(\mu, \nu)^p$.

THEOREM Benamou–Brenier

$$W_p(\mu, \nu)^p = \min \left\{ \int_0^1 \int_{\mathbb{R}^n} |v|^p d\rho_t dt : \partial_t \rho_t + \text{div}(v_t \rho_t) = 0, \rho_0 = \mu, \rho_1 = \nu \right\}.$$

“ $\geq$ ” is the geodesic. “ $\leq$ ”: $|v'| \leq \|v_t\|_{L^p(\rho_t)}$, so $W_p \leq \int_0^1 |v'| \leq \left(\int_0^1 \|v_t\|^p \right)^{1/p}$. In the variables $(\rho, m = v\rho)$ the problem is convex.

Dual of Benamou–Brenier: Hamilton–Jacobi

For $p = 2$, a multiplier $\phi(t, x)$ for the continuity equation gives

$$\frac{1}{2}W_2^2 = \min_{\rho, \nu} \sup_{\phi} \left\{ \int_0^1 \int \left(\frac{1}{2} |v|^2 - \partial_t \phi - \nabla \phi \cdot v \right) d\rho_t dt + \int \phi_1 d\nu - \int \phi_0 d\mu \right\}.$$

Swap **min** and **sup**; minimise in v : $\inf_v \left\{ \frac{1}{2} |v|^2 - \nabla \phi \cdot v \right\} = -\frac{1}{2} |\nabla \phi|^2$ at $v = \nabla \phi$.

Then $\rho \geq 0$ forces

$$\frac{1}{2}W_2(\mu, \nu)^2 = \sup \left\{ \int \phi_1 d\nu - \int \phi_0 d\mu : \partial_t \phi + \frac{1}{2} |\nabla \phi|^2 \leq 0 \right\}.$$

The optimal velocity is a gradient, $v_t = \nabla \phi_t$, and $-\phi_0, \phi_1$ are the Kantorovich potentials.

Wasserstein geometry: convexity and gradient flows



With geodesics in hand we can speak of convex functionals on $\mathcal{P}_2$, prove Brunn–Minkowski, and read classical PDEs as gradient flows.

Displacement convexity

DEFINITION McCann

$F : \mathcal{P}_p(\mathbb{R}^n) \rightarrow \mathbb{R} \cup \{+\infty\}$ is *displacement convex* if $t \mapsto F(\rho_t)$ is convex along every constant-speed geodesic:

$$F(\rho_t) \leq (1 - t)F(\rho_0) + tF(\rho_1).$$

Many energies are not convex for $(1 - t)\mu + t\nu$ but are convex along displacement interpolation. This gives uniqueness of minimisers, functional inequalities, and well-behaved gradient flows.

Three displacement convex functionals

- **Internal energy** $\mathcal{U}(\rho) = \int U(\rho) dx$, if $U(0) = 0$ and $r \mapsto r^n U(r^{-n})$ is convex non-increasing.
 $U = r \log r$ (entropy), $U = \frac{r^m}{m-1}$ with $m \geq 1 - \frac{1}{n}$, $U = -r^{1-1/n}$.
- **Potential energy** $\mathcal{V}(\rho) = \int V d\rho$, V convex.
- **Interaction energy** $\mathcal{W}(\rho) = \frac{1}{2} \iint W(x - y) d\rho d\rho$, W convex.

Why the internal energy: along $\rho_t = (T_t)_\# \mu$ the density is $\mu / \det \nabla T_t$ with $\nabla T_t = (1 - t) \text{Id} + t \nabla^2 \tilde{\varphi} \succeq 0$; $t \mapsto (\det \nabla T_t)^{1/n}$ is concave (Minkowski), and the condition on U turns this into convexity.

THEOREM

For compact $A, B \subset \mathbb{R}^n$ of positive measure and $t \in (0, 1)$,

$$|(1 - t)A + tB|^{1/n} \geq (1 - t) |A|^{1/n} + t |B|^{1/n}.$$

Idea: run $\mathcal{U}(\rho) = - \int \rho^{1-1/n}$ along the geodesic between the uniform measures on A and B .

Setup: $\mu = \frac{1_A}{|A|} dx$, $\nu = \frac{1_B}{|B|} dy$, so $\mathcal{U}(\mu) = -|A|^{1/n}$, $\mathcal{U}(\nu) = -|B|^{1/n}$, and since $T(A) \subset B$,

$$C_t := \text{supp } \rho_t \subset (1 - t)A + tB.$$

1. **Jensen** on C_t (convex U , $\int_{C_t} \rho_t = 1$):

$$\mathcal{U}(\rho_t) \geq |C_t| U(|C_t|^{-1}) = -|C_t|^{1/n} \geq -|(1-t)A + tB|^{1/n}.$$

2. **Displacement convexity**:

$$\mathcal{U}(\rho_t) \leq (1-t)\mathcal{U}(\mu) + t\mathcal{U}(\nu) = -(1-t)|A|^{1/n} - t|B|^{1/n}.$$

Combine the two. ■

Both inequalities rest on **det** ∇T behaving well when T is the gradient of a convex function.

Gradient flows: from $\mathbb{R}^n$ to $\mathcal{P}_2(\mathbb{R}^n)$

In $\mathbb{R}^n$, $\dot{x} = -\nabla f(x)$ is steepest descent. Defining ∇ needs an inner product on velocities.

OTTO'S CALCULUS Velocities at ρ are tangent vectors $\partial_t \rho$; by Benamou–Brenier they have the form

$$\partial_t \rho = -\operatorname{div}(\rho \nabla u), \quad \langle \partial_t \rho^1, \partial_t \rho^2 \rangle_\rho := \int \nabla u_1 \cdot \nabla u_2 \, d\rho.$$

A formal Riemannian metric: the kinetic energy of the velocity field.

The Wasserstein gradient

First variation: $\frac{d}{d\varepsilon} F(\rho + \varepsilon(\tilde{\rho} - \rho)) \big|_0 = \int \frac{\delta F}{\delta \rho} d(\tilde{\rho} - \rho)$. Along a curve with velocity ∇u , the chain rule and the continuity equation give

$$\frac{d}{dt} F(\rho_t) = \int \nabla \frac{\delta F}{\delta \rho} \cdot \nabla u \, d\rho = \langle -\operatorname{div}\left(\rho \nabla \frac{\delta F}{\delta \rho}\right), \partial_t \rho \rangle_\rho.$$

WASSERSTEIN GRADIENT FLOW

$$\partial_t \rho = -\operatorname{grad}_{W_2} F(\rho) = \operatorname{div}\left(\rho \nabla \frac{\delta F}{\delta \rho}(\rho)\right),$$

i.e. particles move with velocity $\mathbf{v} = -\nabla \frac{\delta F}{\delta \rho}$.

energy $F(\rho)$	$\delta F / \delta \rho$	$\partial_t \rho = \operatorname{div}\left(\rho \nabla \frac{\delta F}{\delta \rho}\right)$	
$\int \rho \log \rho$	$\log \rho + 1$	$\partial_t \rho = \Delta \rho$	heat
$\frac{1}{m-1} \int \rho^m$	$\frac{m}{m-1} \rho^{m-1}$	$\partial_t \rho = \Delta \rho^m$	porous medium
$\int \rho \log \rho + \int V \, d\rho$	$\log \rho + 1 + V$	$\partial_t \rho = \Delta \rho + \operatorname{div}(\rho \nabla V)$	Fokker–Planck
$\frac{1}{2} \iint W(x - y) \, d\rho \, d\rho$	$W * \rho$	$\partial_t \rho = \operatorname{div}(\rho \nabla (W * \rho))$	McKean–Vlasov

Jordan–Kinderlehrer–Otto, 1998: the heat equation is the steepest descent of the entropy for W_2 .

Where we stand



All three questions answered. Plans exist; for convex costs they are maps given by a potential; the optimal cost is a distance whose geodesics, convexity and gradient flows recover classical PDE.

The big picture

1. EXISTENCE

Monge's problem is rigid. Kantorovich's plans form a compact convex set: l.s.c. costs have optimal plans.

2. STRUCTURE

Duality: prices $\varphi \oplus \psi \leq c$. Optimal plans live on c -cyclically monotone sets; for convex costs they are maps, for the quadratic cost $T = \nabla(\text{convex})$.

3. GEOMETRY

W_p is a distance. Geodesics move particles on straight lines; displacement convexity; PDEs as gradient flows.

One object, the optimal plan π . Two faces: **where the mass goes**, and **the potential φ** .

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Thank you